

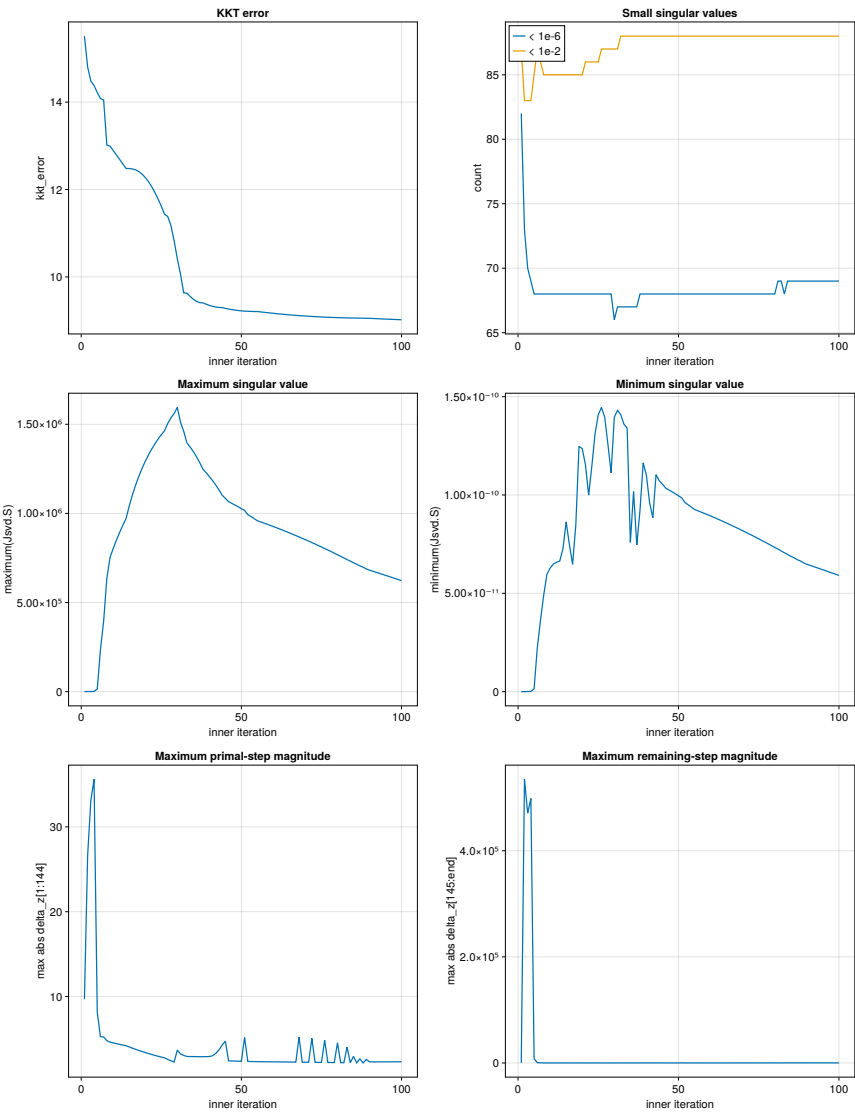
Solver Diagnostics: Naive Solve

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
kkt_error	9.019713e+00	1.550980e+01	1.023334e+01	9.220210e+00
count(Jsvd.S < 1e-6)	6.600000e+01	8.200000e+01	6.832000e+01	6.800000e+01
count(Jsvd.S < 1e-2)	8.300000e+01	8.800000e+01	8.723000e+01	8.800000e+01
maximum(Jsvd.S)	6.939940e+00	1.594583e+06	9.266436e+05	9.088282e+05
minimum(Jsvd.S)	1.329944e-16	1.445105e-10	8.319991e-11	8.302291e-11
maximum(abs.(delta_z[1:144]))	2.168760e+00	3.560762e+01	3.971051e+00	2.621567e+00
maximum(abs.(delta_z[145:end]))	4.640544e+01	5.346457e+05	1.523999e+04	1.087066e+02

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\max \delta z_{145:} $
1	1.550980e+01	82	87	6.939940e+00	1.329944e-16	9.675507e+00	1.817165e+02
2	1.480346e+01	73	83	1.327926e+02	1.115385e-14	2.661907e+01	5.346457e+05
3	1.448053e+01	70	83	7.879145e+02	7.791309e-14	3.312689e+01	4.708610e+05
4	1.438150e+01	69	83	8.726437e+02	8.544994e-14	3.560762e+01	4.985386e+05
5	1.421931e+01	68	85	1.531317e+04	1.513549e-12	8.164553e+00	8.048514e+03
6	1.408236e+01	68	87	2.350397e+05	2.233732e-11	5.261842e+00	4.511482e+02
7	1.404726e+01	68	86	3.968923e+05	3.642801e-11	5.232369e+00	3.642255e+02
8	1.301692e+01	68	85	6.363930e+05	4.914964e-11	4.805826e+00	1.089262e+02
9	1.299140e+01	68	85	7.536476e+05	5.968172e-11	4.622268e+00	1.029960e+02
10	1.289031e+01	68	85	8.039128e+05	6.285065e-11	4.532668e+00	1.003975e+02
11	1.278937e+01	68	85	8.506389e+05	6.488283e-11	4.444571e+00	9.796503e+01
12	1.268782e+01	68	85	8.939971e+05	6.579494e-11	4.358179e+00	9.567036e+01
13	1.258523e+01	68	85	9.342554e+05	6.641492e-11	4.273621e+00	9.349228e+01
14	1.248143e+01	68	85	9.717063e+05	7.274259e-11	4.200179e+00	9.141481e+01
15	1.247924e+01	68	85	1.042866e+06	8.634221e-11	4.054359e+00	8.748436e+01
16	1.247086e+01	68	85	1.105067e+06	7.474760e-11	3.912065e+00	8.384463e+01
17	1.244886e+01	68	85	1.160018e+06	6.474488e-11	3.774518e+00	8.044859e+01
18	1.240970e+01	68	85	1.208994e+06	8.481413e-11	3.642238e+00	7.726340e+01
19	1.235185e+01	68	85	1.252946e+06	1.247223e-10	3.515376e+00	7.426458e+01
20	1.227483e+01	68	85	1.292589e+06	1.235621e-10	3.393892e+00	7.143299e+01
21	1.217880e+01	68	86	1.328471e+06	1.153998e-10	3.277646e+00	6.875306e+01
22	1.206425e+01	68	86	1.361018e+06	9.992882e-11	3.166445e+00	6.621181e+01
23	1.193191e+01	68	86	1.390567e+06	1.147476e-10	3.060072e+00	6.379814e+01
24	1.178261e+01	68	86	1.417388e+06	1.311255e-10	2.958304e+00	6.150246e+01
25	1.161720e+01	68	86	1.441699e+06	1.408086e-10	2.860917e+00	5.931638e+01
26	1.143651e+01	68	87	1.463673e+06	1.445105e-10	2.767691e+00	5.723249e+01
27	1.138311e+01	68	87	1.504272e+06	1.395520e-10	2.590299e+00	5.328733e+01
28	1.117836e+01	68	87	1.536385e+06	1.257939e-10	2.427528e+00	4.969047e+01
29	1.082753e+01	68	87	1.560548e+06	1.113008e-10	2.277765e+00	4.640544e+01
30	1.041359e+01	66	87	1.594583e+06	1.395629e-10	3.683518e+00	9.560402e+01
31	1.006090e+01	67	87	1.511485e+06	1.430213e-10	3.229489e+00	8.338623e+01
32	9.634921e+00	67	88	1.460431e+06	1.408357e-10	3.053302e+00	8.372113e+01
33	9.627376e+00	67	88	1.395484e+06	1.359974e-10	2.933176e+00	8.613031e+01
34	9.552686e+00	67	88	1.372579e+06	1.339744e-10	2.924290e+00	8.741145e+01
35	9.489764e+00	67	88	1.346635e+06	7.592714e-11	2.919691e+00	8.886938e+01
36	9.440924e+00	67	88	1.317300e+06	1.017038e-10	2.913414e+00	9.545032e+01
37	9.410680e+00	67	88	1.284554e+06	7.459350e-11	2.905704e+00	1.018046e+02
38	9.406215e+00	68	88	1.248337e+06	9.214766e-11	2.905301e+00	1.084869e+02
39	9.375648e+00	68	88	1.228236e+06	1.163398e-10	2.913023e+00	1.128141e+02
40	9.349338e+00	68	88	1.206864e+06	1.099417e-10	2.928798e+00	1.180447e+02
41	9.327799e+00	68	88	1.183984e+06	9.598024e-11	3.014332e+00	1.245116e+02
42	9.311599e+00	68	88	1.159264e+06	8.841945e-11	3.295129e+00	1.327184e+02
43	9.301351e+00	68	88	1.132208e+06	1.103199e-10	3.693029e+00	1.433914e+02
44	9.297736e+00	68	88	1.102008e+06	1.072320e-10	4.290157e+00	1.571586e+02
45	9.280484e+00	68	88	1.084653e+06	1.054537e-10	4.727448e+00	1.650058e+02
46	9.264642e+00	68	88	1.065669e+06	1.035060e-10	2.408465e+00	8.209205e+01
47	9.252505e+00	68	88	1.056450e+06	1.025775e-10	2.399098e+00	8.212028e+01
48	9.241580e+00	68	88	1.046925e+06	1.016161e-10	2.389799e+00	8.215038e+01
49	9.231934e+00	68	88	1.037080e+06	1.006204e-10	2.380573e+00	8.218248e+01
50	9.223641e+00	68	88	1.026900e+06	9.958919e-11	2.371422e+00	8.221670e+01
51	9.216778e+00	68	88	1.016370e+06	9.852088e-11	5.163783e+00	1.694087e+02
52	9.214279e+00	68	88	9.936671e+05	9.618560e-11	2.347114e+00	8.216775e+01
53	9.210528e+00	68	88	9.823866e+05	9.503939e-11	2.337974e+00	8.219946e+01
54	9.208445e+00	68	88	9.707092e+05	9.385188e-11	2.328911e+00	8.510462e+01
55	9.208119e+00	68	88	9.586185e+05	9.262159e-11	2.319930e+00	8.816791e+01
56	9.199464e+00	68	88	9.523562e+05	9.198453e-11	2.315478e+00	8.975575e+01
57	9.191033e+00	68	88	9.459826e+05	9.133607e-11	2.311049e+00	9.137861e+01
58	9.182831e+00	68	88	9.394957e+05	9.067602e-11	2.306645e+00	9.303697e+01
59	9.174861e+00	68	88	9.328932e+05	9.000421e-11	2.302265e+00	9.473125e+01
60	9.167129e+00	68	88	9.261730e+05	8.932047e-11	2.297912e+00	9.646186e+01
61	9.159638e+00	68	88	9.193330e+05	8.862462e-11	2.293586e+00	9.822914e+01
62	9.152393e+00	68	88	9.123711e+05	8.791651e-11	2.289289e+00	1.000334e+02

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\max \delta z_{145:} $
63	9.145399e+00	68	88	9.052853e+05	8.719598e-11	2.285022e+00	1.018749e+02
64	9.138659e+00	68	88	8.980736e+05	8.646289e-11	2.280787e+00	1.037537e+02
65	9.132179e+00	68	88	8.907339e+05	8.571711e-11	2.276585e+00	1.056701e+02
66	9.125963e+00	68	88	8.832645e+05	8.495850e-11	2.272418e+00	1.076241e+02
67	9.120014e+00	68	88	8.756634e+05	8.418697e-11	2.268288e+00	1.096155e+02
68	9.114336e+00	68	88	8.679290e+05	8.340242e-11	5.208575e+00	1.611605e+02
69	9.110334e+00	68	88	8.604893e+05	8.264340e-11	2.258603e+00	1.126521e+02
70	9.105025e+00	68	88	8.525569e+05	8.184017e-11	2.254504e+00	1.147214e+02
71	9.099992e+00	68	88	8.444885e+05	8.102391e-11	2.250449e+00	1.168262e+02
72	9.095239e+00	68	88	8.362830e+05	8.019461e-11	5.068211e+00	1.702069e+02
73	9.092130e+00	68	88	8.285234e+05	7.940603e-11	2.240221e+00	1.199824e+02
74	9.087735e+00	68	88	8.201195e+05	7.855874e-11	2.236208e+00	1.221516e+02
75	9.083621e+00	68	88	8.115781e+05	7.769869e-11	2.232250e+00	1.243525e+02
76	9.079789e+00	68	88	8.028988e+05	7.682599e-11	4.833205e+00	1.844947e+02
77	9.077636e+00	68	88	7.948593e+05	7.601324e-11	2.221529e+00	1.275811e+02
78	9.074139e+00	68	88	7.859880e+05	7.512410e-11	2.217636e+00	1.298274e+02
79	9.070919e+00	68	88	7.769815e+05	7.422298e-11	2.213814e+00	1.320999e+02
80	9.067979e+00	68	88	7.678405e+05	7.331015e-11	4.546211e+00	1.987945e+02
81	9.066743e+00	69	88	7.595501e+05	7.247789e-11	2.202684e+00	1.353324e+02
82	9.064097e+00	69	88	7.502332e+05	7.155133e-11	2.198968e+00	1.376262e+02
83	9.061720e+00	68	88	7.407886e+05	7.061424e-11	4.029064e+00	2.087585e+02
84	9.061185e+00	69	88	7.325206e+05	6.978943e-11	2.187854e+00	1.408358e+02
85	9.059068e+00	69	88	7.229174e+05	6.884125e-11	2.946304e+00	2.079167e+02
86	9.058505e+00	69	88	7.152238e+05	6.807741e-11	2.177731e+00	1.440839e+02
87	9.056646e+00	69	88	7.054701e+05	6.711941e-11	2.652836e+00	2.014054e+02
88	9.055454e+00	69	88	6.983579e+05	6.641761e-11	2.168760e+00	1.473440e+02
89	9.053852e+00	69	88	6.884645e+05	6.545129e-11	2.584557e+00	2.016184e+02
90	9.052479e+00	69	88	6.814840e+05	6.476680e-11	2.304260e+00	1.723569e+02
91	9.049021e+00	69	88	6.757172e+05	6.420462e-11	2.293195e+00	1.725385e+02
92	9.045519e+00	69	88	6.699680e+05	6.364547e-11	2.295215e+00	1.745163e+02
93	9.042087e+00	69	88	6.641729e+05	6.308308e-11	2.296039e+00	1.763283e+02
94	9.038713e+00	69	88	6.583394e+05	6.251826e-11	2.297076e+00	1.781698e+02
95	9.035399e+00	69	88	6.524681e+05	6.195113e-11	2.298182e+00	1.800195e+02
96	9.032145e+00	69	88	6.465602e+05	6.138187e-11	2.299372e+00	1.818788e+02
97	9.028949e+00	69	88	6.406174e+05	6.081071e-11	2.300646e+00	1.837466e+02
98	9.025812e+00	69	88	6.346411e+05	6.023787e-11	2.302007e+00	1.856223e+02
99	9.022733e+00	69	88	6.286331e+05	5.966358e-11	2.303454e+00	1.875048e+02
100	9.019713e+00	69	88	6.225953e+05	5.908811e-11	2.304990e+00	1.893932e+02